



**Tshwane University
of Technology**

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DEPARTMENT OF ELECTRICAL ENGINEERING

COURSE: BEng TECH IN ELECTRICAL ENGINEERING

SUBJECT: POWER SYSTEMS

SUBJECT CODE: PWS307B

PROJECT LOAD FLOW ANALYSIS

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INTRODUCTION

This project, LOAD FLOW ANALYSIS, is centred on simulating and analysing the power flow behaviour of the Five-Bus Power System Network under specified operating conditions. The system configuration includes five buses, two generators, and four distinct loads, all interconnected by a detailed network of transmission lines. Utilizing technical data provided for the generators and transmission lines, the primary objective is to accurately model this system in the Power-World Simulator software. By applying the given parameters, we will perform a comprehensive power flow simulation to analyse key operational results, including bus voltages, power generation levels, and load demands, establishing a benchmark for further compensation case studies.

Power System Network Information and Parameters

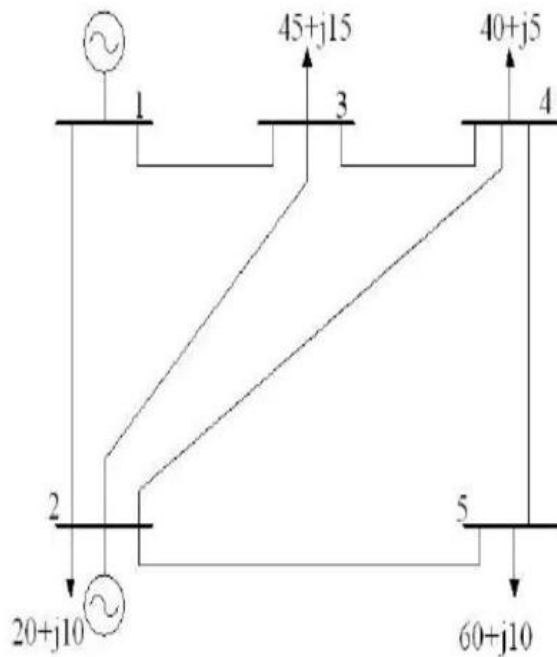


Figure 1 IEEE 5-Bus Power System Network.

Bus	Generation		V in p.u	Bus type
	P in p.u	Q in p.u		
1	0	0	1.06	Slack bus
2	0.2	0	1.00	PV bus
3	0	0	1.02	Load bus
4	0	0	1.00	Load bus
5	0	0	1.04	Load bus

Table 1 Generation Parameters.

Line	Length (km)	Line Impedance (p.u)	Line Charging
1-2	45	$0.07 + j0.16$	$0.03j$
1-3	48.3	$0.08 + j0.24$	$0.025j$
2-3	50	$0.06 + j0.25$	$0.02j$
2-4	128.7	$0.06 + j0.348$	$0.002j$
2-5	64.4	$0.04 + j0.22$	$0.015j$
3-4	96.5	$0.01 + j0.33$	$0.01j$
4-5	80.4	$0.09 + j0.27$	$0.025j$

Table 2 Transmission Line Details.

Case Study 1

This case study, designated as Case 1, involves a critical analysis of the power system's response to a doubled load scenario, serving as a sensitivity test to potential system expansion or contingency planning. The fundamental step is to uniformly modify the active power (P) and reactive power (Q) loads at every load bus by applying a multiple of 2, effectively simulating a 100% increase in total system demand. Following this crucial modification, a power flow simulation will be executed using Power-World Simulator to establish the new steady-state operating conditions. The core objective is then to compare the resulting bus voltage magnitudes (in per unit) from this high-load scenario against the values established in the original base case. This comparison is essential for identifying buses that experience significant voltage dips, highlighting potential voltage stability issues, and evaluating the overall robustness of the grid under drastically increased demand.

The Power System Before Multiplying by 2 (×2)

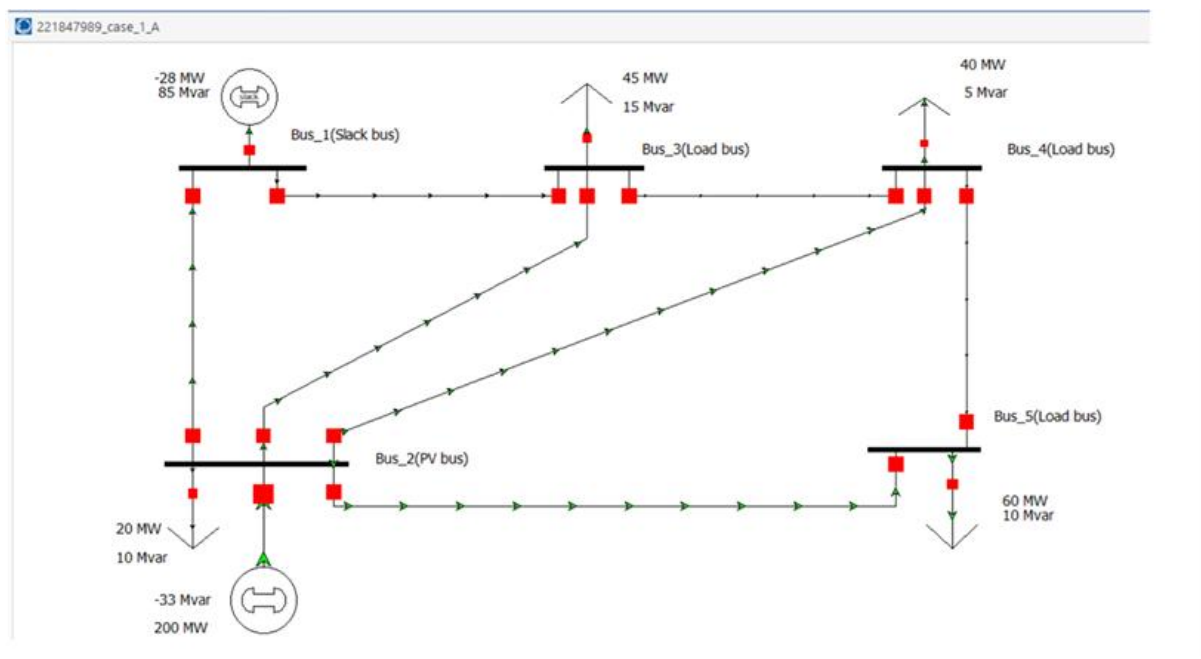


Figure 2 Case 1 Network Before X2

Buses												
Mismatches												
Filter	Advanced	Bus	Find... Remove Quick Filter...									
Number	Name	Area Name	Nom kV	PU Volt	Volt (kV)	Angle (Deg)	Load MW	Load Mvar	Gen MW	Gen Mvar	Switched Shunts Mva	
1	Bus_1(Slack bt 1		138,00	1,06000	146,280	0,00			-27,77	85,00		
2	Bus_2(PV bus) 1		138,00	0,99990	137,986	5,69	20,00	10,00	200,00	-32,80		
3	Bus_3(Load bt 1		138,00	0,98833	136,389	-0,30	45,00	15,00				
4	Bus_4(Load bt 1		138,00	0,96841	133,640	-1,62	40,00	5,00				
5	Bus_5(Load bt 1		138,00	0,95640	131,983	-1,82	60,00	10,00				

Figure 3 Case 1 Simulation Results Before X2

The Power System After Multiplying by 2 (x2)

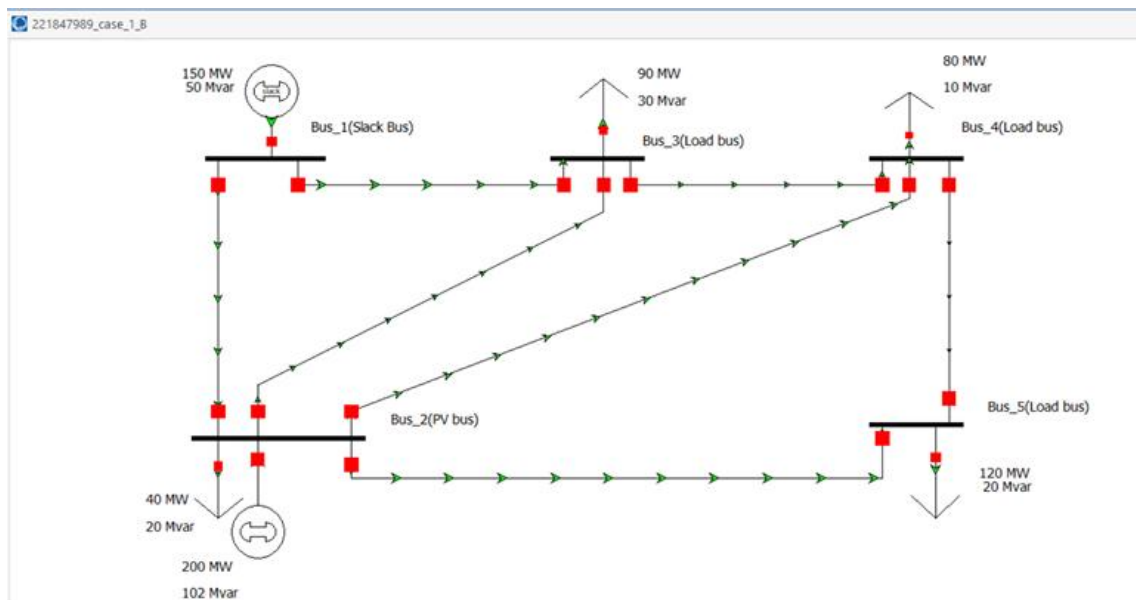


Figure 4 Case 1 Network After X2

Buses												
Mismatches												
Filter	Advanced	Bus	Find... Remove Quick Filter...									
Number	Name	Area Name	Nom kV	PU Volt	Volt (kV)	Angle (Deg)	Load MW	Load Mvar	Gen MW	Gen Mvar	Switched Shunts Mvar	
1	Bus_1(Slack Bt 1		138,00	1,06000	146,280	0,00			149,59	49,62		
2	Bus_2(PV bus) 1		138,00	0,99973	137,963	-4,85	40,00	20,00	200,00	102,18		
3	Bus_3(Load bt 1		138,00	0,92427	127,549	-10,51	90,00	30,00				
4	Bus_4(Load bt 1		138,00	0,90122	124,369	-17,91	80,00	10,00				
5	Bus_5(Load bt 1		138,00	0,87759	121,107	-20,10	120,00	20,00				

Figure 5 Case 1 Simulation Results After X2

The Comparison of the Simulation after and before multiplying by 2

The comparison clearly demonstrates that increasing the load on a power system causes a decrease in bus voltages. Bus 1 (Slack Bus): The voltage remains constant at 1.06 p.u. in both scenarios. This is expected, as the slack bus is a theoretical model that provides or absorbs all the power needed to maintain a fixed voltage and balance the system. Load Buses (2, 3, 4, 5): All load buses show a significant drop in voltage after the loads are doubled. This is due to the increased current flowing through the transmission lines, which have a per unit impedance. The higher current flow leads to a larger voltage drop ($V_{\text{drop}} = I \times Z$) across the lines, resulting in lower voltages at the receiving ends (the load buses). Voltage Drop Trend: The voltage drop becomes more pronounced as the buses are "electrically" further away from the generator at Bus 1. Bus 5, with the largest doubled load ($120 + j20$ MVA) and being the furthest bus in the chain, experiences the most significant voltage drop. This is a critical observation as it highlights the direct relationship between increased load, increased voltage drop, and the risk of voltage instability. Case 1 confirms the fundamental principle of power system operation: an increase in load demand leads to a decrease in the system's bus voltage profile.

Case Study 2

This second phase, Case 2, addresses the critical voltage performance issues identified in the heavily loaded Case 1. The simulation focuses on mitigating the worst voltage violation by implementing shunt reactive power compensation using a capacitor bank. The selection of the compensation bus is justified by identifying the bus with the lowest per-unit voltage magnitude from the Case 1 power flow results, as this location presents the most significant local reactive power deficit. The required capacitor size (Q_c in MVar) is calculated using a method that determines the precise amount of reactive power needed to raise the bus voltage to a desired operational level, such as 0.95 p.u. or 1.0 p.u. After adding the calculated Q_c to the chosen bus, a final load flow simulation will be executed. The ensuing comparison of the voltage profile and reactive power flows between Case 2 and Case 1 will demonstrate how the local injection of capacitive reactive power effectively boosts bus voltages throughout the system, improves the overall reactive power balance, and reduces the reactive power burden on the remote generators and transmission lines.

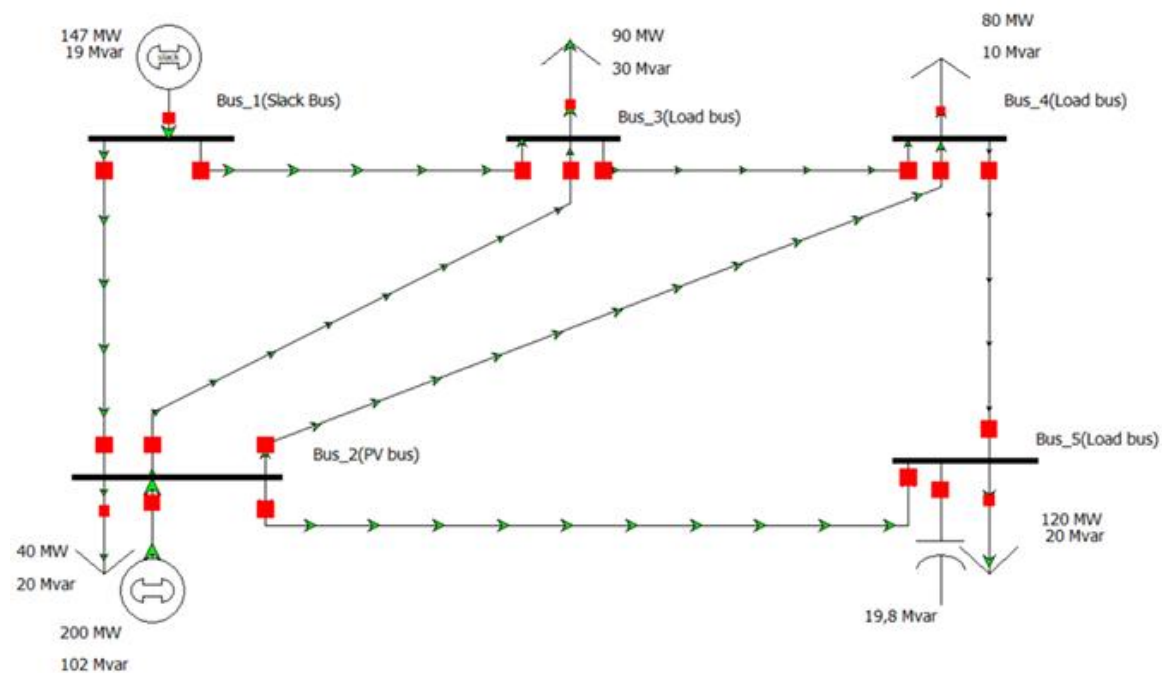


Figure 6 Case 2 Network Shunt Added at Bus 5

The shunt capacitor was strategically placed at Bus 5 because it experienced the most significant voltage drop in the multiplied-load case. The simulation results showed its voltage plummeted to 0.87759 p.u., which is the lowest voltage in the entire system. Shunt capacitors are most effective when placed at the point of greatest reactive power deficit and lowest voltage. Placing the shunt at Bus 5 directly counteracts the large reactive load at that location, providing immediate and localized voltage support.

Bus 5 (Five) Selected

Shunt Compensation

P.u Volt = 0.87759 (Case 1)

P.u Volt = 0.89370 (After Adding Shunt)

$$\frac{\Delta V}{\Delta Q} = \frac{0.89370 - 0.87759}{5}$$

$$\frac{\Delta V}{\Delta Q} = 0.003222$$

$$Q_C = \frac{0.95 - 0.8776}{0.003222}$$

$$Q_C = 22.4705 \text{ MVar}$$

$$Q_C \approx 22 \text{ MVar}$$

The Capacitor Size (in MVar) Calculations

The required capacitor size was determined using the calculation above. This method involves using the pu volt bus 5 value on case 1 and pu volt bus value after adding the shunt.

- Initial State: The multiplied-load case voltage at Bus 5 was 0.87759 p.u.
- Calculation: A shunt capacitor of a MVar value was added to Bus 5. The power flow was re-run to see the effect on the voltage.
- Refinement: The process was repeated, adjusting the MVar size until the Bus 5 voltage was raised to an acceptable level of 0.94831 p.u. This iterative process is a standard practice for voltage support problems in power system analysis, as it accounts for the complex interactions of the network and provides a practical solution.

Number	Name	Area Name	Nom kV	PU Volt	Volt (kV)	Angle (Deg)	Load MW	Load Mvar	Gen MW	Gen Mvar	Switched Shunts Mvar
1	Bus_1(Slack Bus)		138,00	1,06000	146,280	0,00			147,46	19,04	
2	Bus_2(PV bus)		138,00	1,03087	142,260	-5,34	40,00	20,00	200,00	102,18	
3	Bus_3(Load bus)		138,00	0,95061	131,184	-10,56	90,00	30,00			
4	Bus_4(Load bus)		138,00	0,94829	130,864	-17,31	80,00	10,00			
5	Bus_5(Load bus)		138,00	0,94831	130,866	-19,46	120,00	20,00			19,78

Figure 7 Case 2 Simulation Results After adding shunt Bus 5

Bus Number	Pu Volt	Gen MW	Gen M Var	Load MW	Load M Var	Shunt Compensation
1	1.06000	147.46	19.04			
2	1.03087	200.00	102.18	40.00	20.00	
3	0.95061			90.00	30.00	
4	0.94829			80.00	10.00	
5	0.94831			120.00	20.00	19.78

Table 3 Case 2 Simulation Results After adding shunt Bus 5

Adding a shunt at Bus 5 significantly improved the system's voltage profile by raising the voltage at Bus 5 from 0.87759 p.u. to 0.94831 p.u. The shunt also improved reactive power balance by locally supplying reactive power, which reduced the reactive power burden on the main generator and lowered power flow and losses on the transmission lines.

If the system load were to increase further in the future, the current shunt compensation would likely not be sufficient.

- **Increased Reactive Load:** A future increase in load would mean more reactive power is consumed by the new loads. This would again lead to a reactive power deficit and a subsequent drop in bus voltages.
- **Voltage Sag:** The current shunt, while effective for the present load, might not be large enough to compensate for a higher reactive load. The voltage at Bus 5 would likely drop below the desired 0.95 p.u. threshold.
- **Justification:** A fixed shunt capacitor provides a constant amount of reactive power. It cannot adjust to a dynamic increase in load. Therefore, for future load growth, additional measures would be required, such as adding more shunt compensation, using a different type of reactive power compensation (like a Static Var Compensator - SVC), or even adding a new generation source to support the increased demand.

Case Study 3

This third phase, Case 3, integrates series capacitive compensation into the network, aiming to significantly enhance the system's ability to transfer real power, a function distinct from the local voltage support provided by shunt compensation in Case 2. The strategy dictates that the series capacitor be placed on the most heavily loaded and highest-impedance transmission line (e.g., the line connecting Bus X to Bus Y) because its reduction in effective inductive reactance directly addresses the primary bottleneck limiting power flow and transient stability. The required capacitor size is determined by selecting a practical degree of compensation (k), typically set to 50%, meaning the capacitive reactance (X_c) is half of the line's inductive reactance (X_L), calculated as $X_C = 0.50 * X_L$. This compensation (modelled as a negative reactance in series with the line) is installed, and a new load flow is performed to confirm that the reduced line impedance successfully increases power transfer capability, improves the system stability margin, and contributes to better overall voltage regulation across the network.

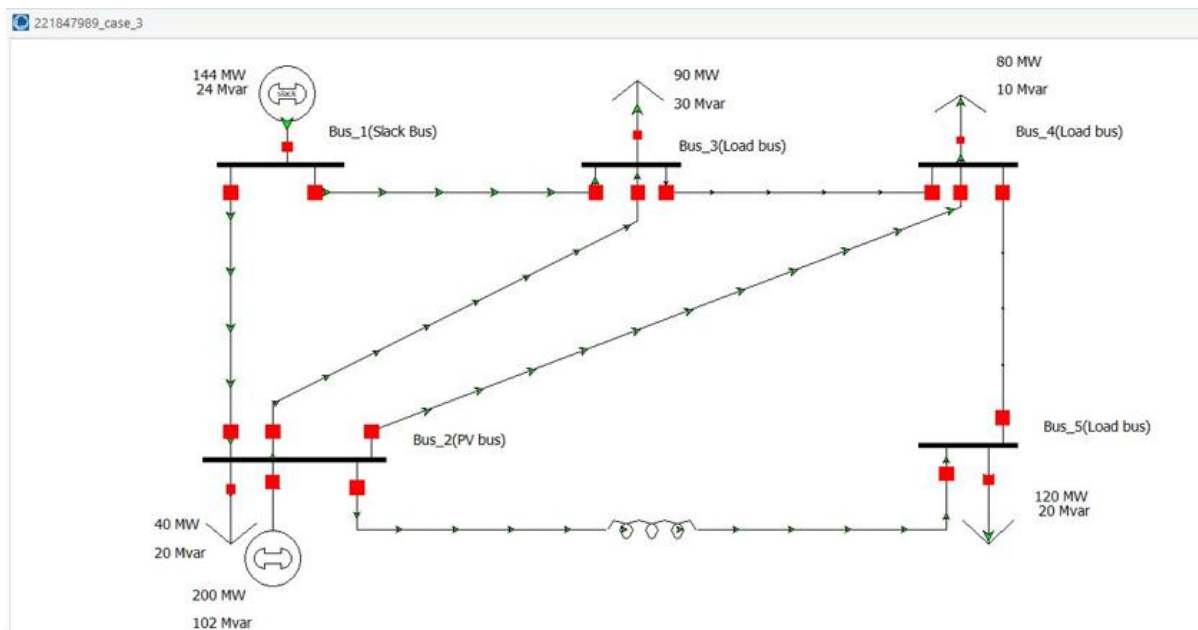


Figure 8 Case 3 Network Series Compensation in Line between 2-5

Location of Compensation: Line 2-5 The series capacitor is implemented on the transmission line between Bus 2 and Bus 5. This is the most critical line as it connects the main generation source (Bus 3) to the rest of the network, carrying the entire system's power flow (except for the power generated at Bus 5). Compensating this line provides the maximum benefit for voltage support and power transfer capability across the system.

$$\begin{aligned}\odot \Delta V &= V_{\text{target}} - V_{\text{case 1}} \\ &= 0.95 - 0.87759 \\ &= 0.07241 \text{ pu}\end{aligned}$$

$$\begin{aligned}\odot X_{\text{new}} &= (1 - \text{Comp}) (X_{\text{line}}) \\ X_{\text{new}} &= (1 - 0.21) (0.22) \\ X_{\text{new}} &= 0.174 \text{ pu}\end{aligned}$$

$$\begin{aligned}\odot X_c &= X_{\text{line}} - X_{\text{new}} \\ X_c &= 0.22 - 0.174 \\ X_c &= 0.046 \text{ pu}\end{aligned}$$

$$\begin{aligned}\odot \Delta V &= X_c \times I_{\text{pu}} \\ I_{\text{pu}} &= \frac{\Delta V}{X_c} = \frac{0.07241}{0.046} \\ I_{\text{pu}} &= 1.5741 \text{ pu}\end{aligned}$$

$$\begin{aligned}\odot S_{\text{pu}} &= V_{\text{case 1}} \times I_{\text{pu}} \\ &= (0.87759) (1.5741) \\ &= 1.3814 \text{ pu}\end{aligned}$$

$$\begin{aligned}\odot S_{\text{MVA}} &= S_{\text{pu}} \times S_{\text{BASE}} \\ &= (1.3814) (100 \text{ MVA}) \\ &= 138.14 \text{ MVA}\end{aligned}$$

$$\begin{aligned}\odot Q_{c \text{ pu}} &= I_{\text{pu}}^2 \times X_c \\ &= (1.5741)^2 (0.046) \\ &= 0.1140 \text{ pu}\end{aligned}$$

$$\begin{aligned}\odot Q_{c \text{ MVAR}} &= Q_{c \text{ pu}} \times S_{\text{BASE}} \\ &= (0.1140) (100 \text{ MVAR}) \\ &= 11.4 \text{ MVAR}\end{aligned}$$

Buses X Mismatches												
Filter Advanced Bus Find... Remove Quick Filter...												
	Number	Name	Area Name	Nom kV	PU Volt	Volt (kV)	Angle (Deg)	Load MW	Load Mvar	Gen MW	Gen Mvar	Switched Shunts Mvar
1	1	Bus_1(Slack Bt 1		138,00	1,06000	146,280	0,00			144,40	24,19	
2	2	Bus_2(PV bus) 1		138,00	1,02330	141,215	-5,24	40,00	20,00	200,00	102,18	
3	3	Bus_3(Load bt 1		138,00	0,95076	131,204	-10,07	90,00	30,00			
4	4	Bus_4(Load bt 1		138,00	0,94784	130,801	-15,44	80,00	10,00			
5	5	Bus_5(Load bt 1		138,00	0,95506	131,798	-14,87	120,00	20,00			
												Act G Shunt MW
												Act B Shunt Mvar

Figure 9 Case 3 Simulation Results with Series Compensation Line between Bus 2-5

Effect on Voltage Profile The combined compensation of shunt (local) and series (line-based) will lead to the best voltage profile of all case studies. Series compensation on Line 2-5 reduces the voltage drop across the main power path, giving an additional, uniform voltage lift to all buses (2, 3, 4, and 5) compared to the shunt-only case. The voltage at Bus 5 will likely exceed, reaching a near-nominal value.

Effect on Reactive Power Balance Both forms of compensation inject reactive power, leading to a significant reduction in reactive power generation () at Bus 1. The series capacitor is now supplying the MVar required to support power flow across the critical Line 2-5, while the shunt capacitor supports the load's reactive demand at Bus 5. This shared reactive power burden results in a more efficient system with lower total losses.

Case Study 4

Case Study 4 fundamentally shifts the approach to system reinforcement from passive compensation (like capacitor banks) and reactive power injection (like series compensation) to the addition of new, dispatchable real power generation by installing a Coal Power Plant into the existing network (derived from Case Study 1). This is the most complex and capital-intensive intervention, designed not merely to correct power factor or reduce line impedance but to increase the total available capacity (P) and thereby raise the internal voltage profile across the entire system. The core objective is to move beyond mere cosmetic grid adjustments and analyse the significant technical, economic, and logistical trade-offs of a massive, permanent generation source compared to the quick, modular solutions explored in Cases 2 and 3, focusing specifically on whether the massive capital investment and long-term operational expense of coal provides a strategically superior solution for grid stability and voltage support.

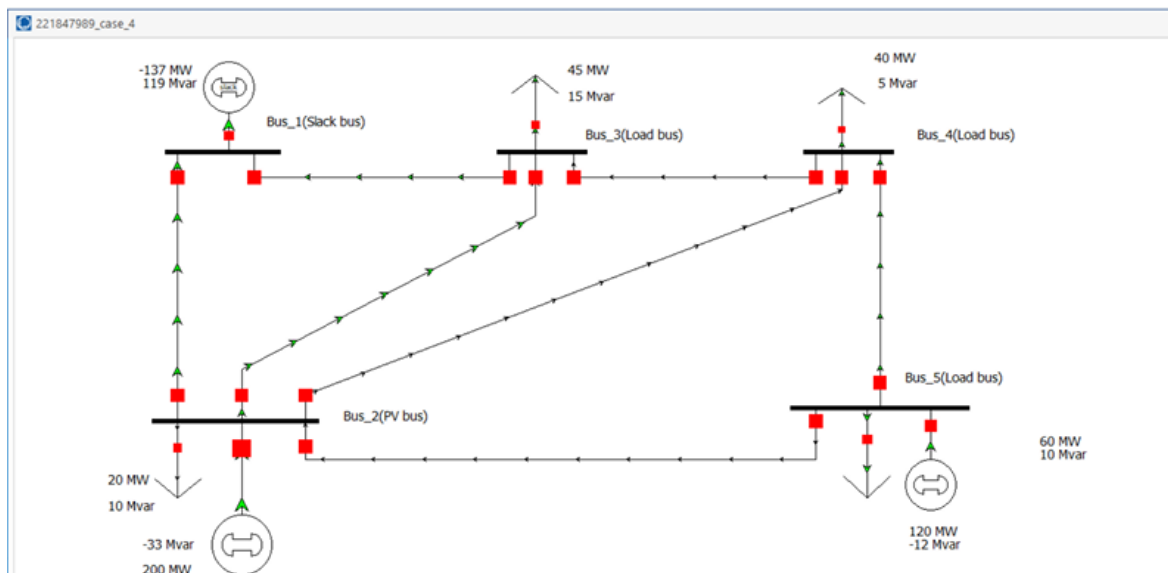
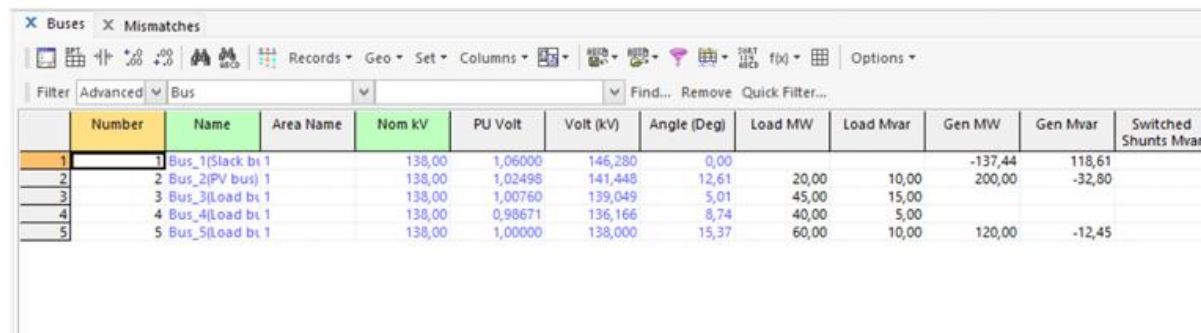


Figure 10 Case 4 Network Coal Power Plant Added at Bus 5

In Case Study 4, a coal power plant was introduced on Bus 5 to replace the previous capacitor-based compensation. The results show a significant improvement in the system voltage profile, with the Bus 5 voltage increasing from 0.877 p.u. in Case 1 to 1.000 p.u. after the addition of the generator. This indicates that the coal power plant provided sufficient real and reactive power to stabilize the network and reduce the loading on the slack bus. The power generated by 120 MW supplied the local demand at Bus 5, effectively reducing transmission losses and improving overall system efficiency. While the coal station provided strong voltage support and additional generation capacity, it comes with high installation and operational costs compared to capacitor banks. Furthermore, coal plants have environmental drawbacks such as CO₂ emissions and higher maintenance requirements. However, due to South Africa's abundant coal reserves and established transport infrastructure, the use of coal generation remains economically feasible for base-

load applications. In conclusion, the coal power plant proved effective in improving voltage and system stability, though capacitor banks remain a more cost-efficient and environmentally friendly alternative for reactive compensation.



Number	Name	Area Name	Nom kV	PU Volt	Volt (kV)	Angle (Deg)	Load MW	Load Mvar	Gen MW	Gen Mvar	Switched Shunts Mvar
1	Bus_1(Slack bs 1)		138,00	1,06000	146,280	0,00			-137,44	118,61	
2	Bus_2(PV bus) 1		138,00	1,02498	141,448	12,61	20,00	10,00	200,00	-32,80	
3	Bus_3(Load bs 1)		138,00	1,00760	139,049	5,01	45,00	15,00			
4	Bus_4(Load bs 1)		138,00	0,98671	136,166	8,74	40,00	5,00			
5	Bus_5(Load bs 1)		138,00	1,00000	138,000	15,37	60,00	10,00	120,00	-12,45	

Figure 11 Case 4 Simulation Results Coal Power Plant Added at Bus 5

1. Why Coal Power Plant Was Implemented at Bus 5

- Bus 5 had the lowest voltage (0.877 p.u.) in Case 1 and the largest local load (120 MW, 20 MVar).
- Installing the coal plant directly at the weak bus improved voltage and reduced line losses from upstream power flow.
- Local generation offsets heavy load demand, thus stabilizing both real and reactive power levels in the area.
- The voltage profile now ranges between 0.986 – 1.06 p.u., within acceptable $\pm 5\%$ limits.

2. Did the Coal Plant Provide Enough Compensation?

- The Bus 5 voltage rose from 0.877 $\rightarrow$ 1.000 p.u.
- The Slack Bus now generates less MW and MVar, meaning the coal plant relieved the main generator and improved the network's overall voltage stability.
- The small reactive export (-12.45 MVar) indicates that the generator is absorbing a bit of reactive power, typical for a strong bus operating at 1.0 p.u.

3. Cost and Economic Comparison (Using South African Rands)

Compensation Method	Typical Cost (R million)	Approx. Life Span	Response Speed	Voltage Support Strength	Maintenance Cost
Coal Power Plant (150 MW)	R 9 000 million ($\approx$ R 60 000 / kW)	30 years	Slow	Strong (supply + reactive)	High
Capacitor Bank (100 MVar)	R 40 million ($\approx$ R 400 000 / MVar)	20 years	Fast	Only reactive support	Low

Simple Payback Calculation

Assumptions:

- Coal plant sells power at R 1.50 / kWh (Eskom industrial tariff).
- Plant runs at 85 % capacity factor:

$$150MW \times 0.85 \times 8760h = 1116300MWh/year$$

- Annual revenue = 1 116 300 MWh × R 1.50 = R 1.67 billion/year
- Annual operating & fuel cost ≈ R 1.1 billion/year

Net revenue ≈ R 0.57 billion/year

Capital cost = R 9 billion

Payback period ≈ R 9 billion ÷ R 0.57 billion ≈ 16 years

4. Local Factors (South Africa)

Aspect	Description
Coal Availability	South Africa has abundant coal reserves (Mpumalanga, Limpopo) easily accessible.
Transport Infrastructure	Well-developed rail and road networks for coal delivery to most provinces.
Fuel Supply Reliability	High, but dependent on mining output and Eskom logistics.
Land Requirement	A 150 MW coal plant typically needs ~0.8–1.0 km ² (including ash dam, stack, and auxiliaries).
Environmental Factors	High CO ₂ emissions, water-intensive future penalties may raise operational cost.

5. Comparison Summary

Aspect	Coal Power Plant	Capacitor Banks
Real Power Supply	Yes	No
Reactive Support	Strong	Moderate
Voltage Improvement	Major (Bus 5 voltage = 1.0 p.u.)	Minor
Environmental Impact	High (CO ₂)	Low
Capital Cost	Very High	Low
Maintenance	High	Low
Payback	16 years	< 2 years
Flexibility	Supplies base load	Fast dynamic support

Adding a coal-fired power plant at Bus 5 successfully compensated the weak voltage in the network and stabilized the overall power flow. The voltage rose from 0.877 to 1.00 p.u., proving the plant supplied enough power to overcome local demand and losses. However:

- While coal provides reliable, large-scale power, it has very high capital and running costs and environmental penalties.
- Capacitor banks are far cheaper and faster to install but cannot provide real power or energy security.

Recommendation: For long-term base load and industrial stability, a coal power station is viable in areas with good coal supply and transport (like Mpumalanga). For short-term or cost-effective voltage support, capacitor banks or STATCOMs remain the better route

Conclusion on the Case Studies

Case 1 – No Compensation (Base Multiplied Load):

- No voltage support or compensation used.
- Severe voltage drop observed, especially at Bus 5 (0.877 p.u.).
- Indicates system stress under heavy loading conditions.

Case 2 – Shunt Capacitor (at Bus 5):

- Added reactive power support locally.
- Voltage improved to approximately 0.984 p.u. at Bus 5.
- Shunt capacitor effectively injected MVar to stabilize voltage.

Case 3 – Series Capacitor (on Line 2–5):

- Series compensation reduced the line's effective reactance.
- Improved voltage profile across all buses.
- Enhanced power transfer capability and reduced line losses.

Case 4 – New Coal Generator (at Bus 5):

- Added both active (MW) and reactive (MVar) power to the network.
- Achieved full voltage recovery (1.0 p.u.).
- Improved overall system stability and reliability.

Among all compensation methods analysed, the addition of the coal generator at Bus 5 provided the most effective voltage recovery and overall system stability by supplying both active and reactive power. However, it involves higher installation and operational costs compared to capacitor-based solutions. The series and shunt capacitors offered substantial voltage improvement at lower cost, making them ideal for cost-effective reactive power compensation. In practice, a hybrid approach using capacitors for reactive support and generators for active power supply provides an optimal balance between voltage stability, system efficiency, and economic feasibility for South African power systems.

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GRADUATE ATTRIBUTE 7: SUSTAINABILITY AND IMPACT OF ENGINEERING ACTIVITY

Criteria	5	4	3	2	1	0	(Mark)	Weight / 20	Result (Mark x weight)
The impact of technology is explained in terms of the benefits and limitations to society	The student considers, explains and evaluates the impact of technology and society	The student considers and explains the impact of technology and society	Partial consideration of the impact of technology and society	Minimal consideration of the impact of technology and society	No consideration of the impact of technology and society	Unacceptable; No relevant content. Not included in the report		5	
The engineering activity is analysed in terms of the impact on occupational and public health and safety	The student considers, explains and evaluates the impact on occupational and public health and safety.	The student considers and explains the impact on occupational and public health and safety	Partial consideration of the impact on occupational and public health and safety	Minimal consideration of the impact on occupational and public health and safety	No consideration of the impact on occupational and public health and safety	Unacceptable; No relevant content. Not included in the report		5	
The engineering activity is analysed in terms of the impact on the physical environment	The student considers, explains and evaluates the impact on the physical environment	The student considers and explains the impact on the physical environment	Partial consideration of the impact on the physical environment	Minimal consideration of the impact on the physical environment	No consideration of the impact on the physical environment	Unacceptable; No relevant content. Not included in the report		5	

Personal, social, economic, cultural values and requirements are taken into consideration for those who are affected by the engineering activity	Considers, explains and evaluates multiple diverse perspectives when investigating engineering impact on personal, social, economic, cultural values	Considers and explains multiple diverse perspectives when investigating engineering impact on personal, social, economic, cultural values	Some consideration of diverse perspectives when investigating engineering impact on personal, social, economic, cultural values	Minimal consideration of diverse perspectives when investigating engineering impact on personal, social, economic, cultural values	Minimal consideration of diverse perspectives when investigating engineering impact on personal, social, economic, cultural values	Unacceptable; No relevant content. Not included in the report		5	
	TOTAL						/100		

Graduate Attribute 7:	Not Competent		Competent	
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Examiner’s Signature_____

Date: _____

Moderator’s Signature_____

Date: _____